

# Can question predictability reveal benchmark exposure?

Three experiments comparing an OpenWebText-trained GPT-2 with MiniCPM5 Base. We tested benchmark familiarity against controls, asked whether familiarity changed with benchmark age, and repeated the primary comparisons with calibration-free word ranks.

## Research question

If a model encountered benchmark questions during training, it might predict their exact wording unusually well. We asked whether question-continuation loss could reveal that familiarity, and whether the newer model showed a different pattern from the older model. Lower loss alone cannot distinguish exposure from stronger general language prediction.

## Shared methodology: predict the question

**We scored question text, excluding answer options and correct answers.** For BoolQ and SocialQA, we concatenated the context/passage and question. The prefix boundary applies to this combined text, so some scored suffix tokens are context rather than the question itself. Other tasks use their question stem or completion context.

Each model received the first 25%, 50% or 75% of the words, with a token-boundary adjustment, and scored the remaining text. The primary setting was **50% prefix with no benchmark cue**. Scoring was teacher forced: for every next token, the model saw all preceding true tokens. We measured how much probability it assigned to the actual continuation; we did not ask it to generate a complete question freely.

**Loss:** sum of negative log probabilities, converted to bits, divided by the UTF-8 byte coverage of scored tokens. We averaged this bits/byte value across items. Lower values indicate easier prediction. Boundary-crossing tokens were excluded; the two tokenizers may therefore score slightly different suffix boundaries.

**Models:** Stanford GPT-2 Small (124M parameters), documented as trained on OpenWebText, and MiniCPM5-1B-Base (1.08B parameters), whose pretraining mixture includes Ultra-FineWeb and other datasets. Both are base checkpoints. Size, architecture, tokenizer and training mixture differ, so this is not a controlled comparison of the two corpora alone.

Report order: Experiment 1 design and controls; Experiment 1 findings; Experiment 2 temporal hypothesis and loss slopes; Experiment 3 word-rank robustness; interpretation and reproducibility.

# Experiment 1 / Design and controls

**Question:** does OLMES wording look more like familiar corpus text than a benchmark released after both checkpoints? Under the exposure hypothesis, OLMES should show lower continuation loss relative to appropriately matched unseen text, including after descriptive corpus normalization.

**OLMES:** 30 items, three from each of ten components: ARC Easy, ARC Challenge, BoolQ, CommonsenseQA, HellaSwag, OpenBookQA, PIQA, SocialIQA, WinoGrande and MMLU. Selection used a frozen random seed. We sampled component evaluation splits rather than reproducing the official capped OLMES evaluation ordering.

**Corpus controls:** 30 OpenWebText passages and 30 Ultra-FineWeb passages, sampled from saved corpus records. Excerpt lengths were matched to the pilot OLMES word lengths with a 12-100-word bound. These are verified corpus members, but we have not verified that the exact records were consumed by the selected model checkpoints. They are not gold-standard training positives.

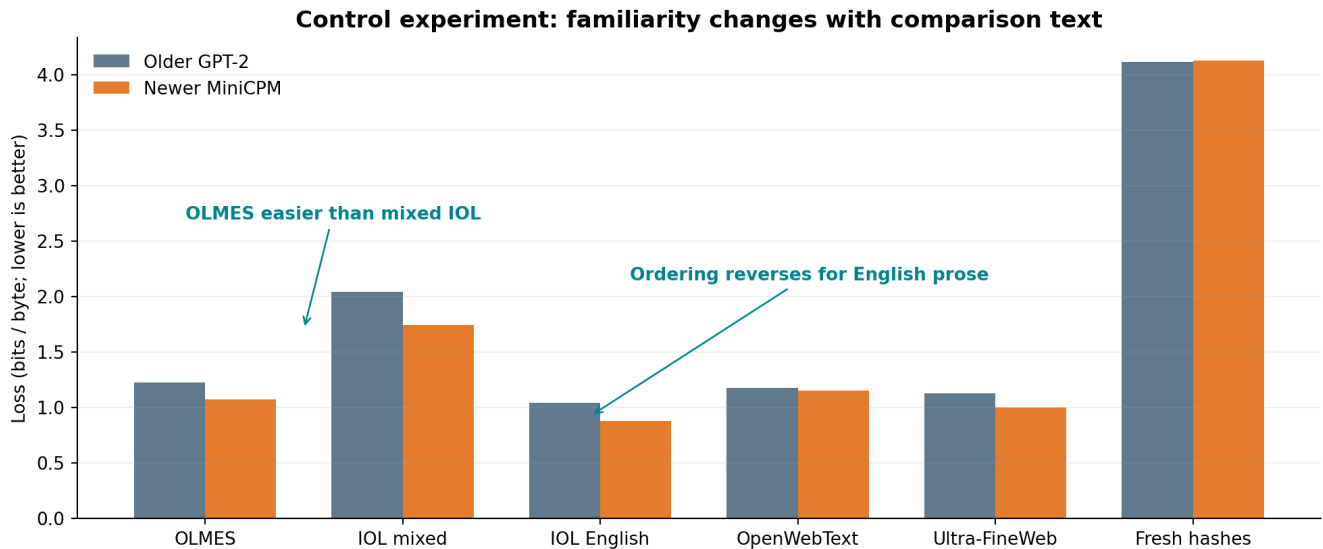
**Post-checkpoint benchmark:** 30 non-overlapping mixed-text excerpts from five human-composed IOL 2026 individual contest problems, six excerpts per problem. The contest followed both pinned checkpoints. Excerpts include unfamiliar languages and linguistic tables; they are not 30 independent questions. Text extraction does not reproduce diagrams.

**English-prose sensitivity:** ten short passages from the same five IOL problems, selected by textual role rather than model score. This exploratory addition addressed the mismatch in language and format. It provides only five independent problem clusters and shares source material with the mixed set. Underlying facts or familiar wording may predate the contest.

**Randomness control:** 30 fresh SHA-256 digests of cryptographically random bytes. These were generated after both checkpoints and frozen for reproduction. They test prediction of unpredictable strings; they are not a natural-language nonmember baseline.

The final pilot contains **160 texts**: 30 OLMES + 60 corpus passages + 30 mixed IOL excerpts + 10 English IOL excerpts + 30 hashes. IOL public provenance supports human authorship, but does not establish that every fragment was previously unseen or exclude private drafts.

# Experiment 1 / What we observed



**Expected:** lower loss on exposed OLMES wording than on suitably matched novel benchmark text.

**Observed:** OLMES was easier than mixed IOL material, but the mean ordering reversed for English-only IOL prose.

For GPT-2, mean loss was 1.220 bits/byte on OLMES, 2.043 on mixed IOL and 1.041 on English IOL. MiniCPM showed the same ordering: 1.069, 1.742 and 0.874 respectively.

For mixed IOL minus OLMES, the differences were **+0.822 [0.649, 0.993]** for GPT-2 and **+0.673 [0.502, 0.840]** for MiniCPM. For English IOL minus OLMES, they were **-0.179 [-0.497, 0.125]** and **-0.195 [-0.464, 0.070]**. The English-prose intervals include zero.

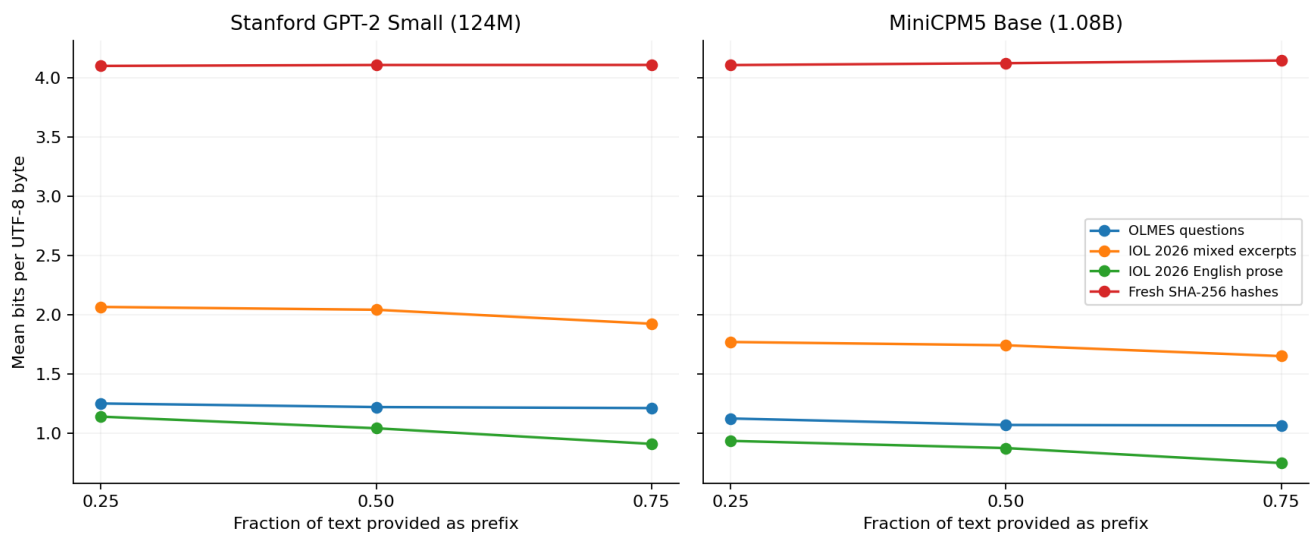
Intervals use 5,000 bootstrap draws over document/problem clusters. With only five IOL problems, they are exploratory and unstable. They do not correct for the differences between benchmark domains.

**Interpretation:** the apparent OLMES familiarity advantage depends strongly on the comparison text. This result does not establish benchmark contamination.

# Experiment 1 / Calibration and sensitivity

**Normalization:** for each model and prefix fraction, we computed  $z = (\text{item loss} - \text{mean own-corpus-control loss}) / \text{own-corpus-control standard deviation}$ . GPT-2 used OpenWebText controls and MiniCPM used Ultra-FineWeb controls. Both prompt conditions used the neutral corpus baseline. This is a descriptive scale, not a training-membership probability.

At the half-text boundary, OLMES z scores were +0.165 for GPT-2 and +0.345 for MiniCPM; mixed IOL scored +3.034 and +3.594; English IOL scored -0.459 and -0.598. Thus normalization retained the same control-dependent ordering. Different model-specific standard deviations limit comparisons between their z scores.



**Prefix sensitivity:** we also scored continuations after 25% and 75% prefixes. The chart shows the secondary descriptive results. Changing the prefix changes both available context and the target suffix; it does not isolate memorization.

**Benchmark cue:** the OLMES cue was a benchmark label followed by “Question:”. Cued-minus-neutral mean loss at 50% was +0.023 bits/byte for GPT-2 and -0.003 for MiniCPM. There was no large descriptive cue benefit; no formal cue-effect test was prespecified.

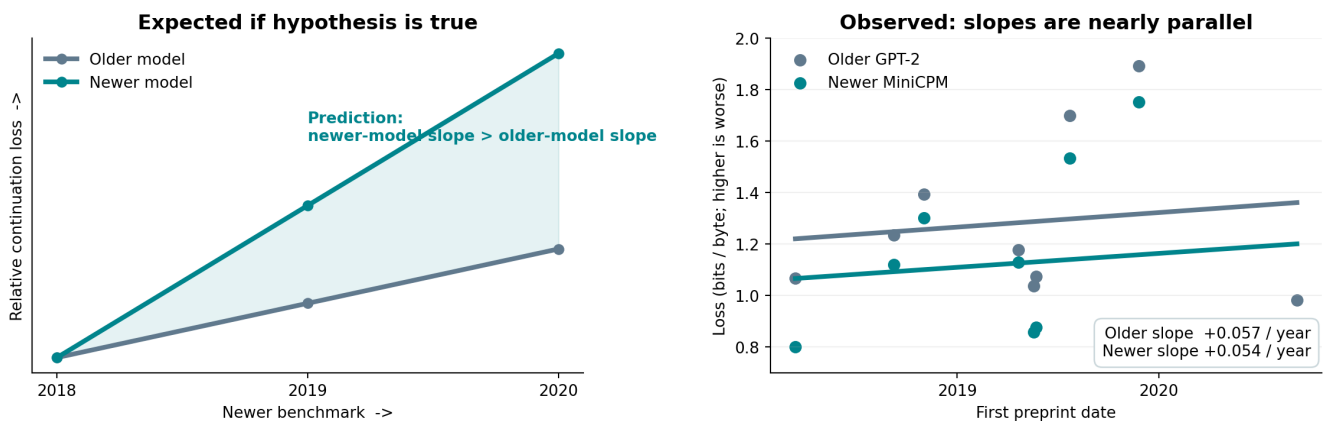
**Fresh hashes:** losses were 4.111 and 4.126 bits/byte, close to the 4-bit ideal per random hexadecimal character. This checks an unpredictable-text baseline. Token accuracy and worst-20%-token loss are secondary diagnostics; token accuracies across different vocabularies are not directly comparable.

## Experiment 2 / Hypothesis and design

The control mismatch in Experiment 1 motivated a second comparison within OLMES. We asked whether the newer model loses more of its prediction advantage as benchmark date increases.

**Specific hypothesis:** if the newer model benefits disproportionately from exposure to older benchmark wording, and that benefit diminishes on later benchmarks, its loss should increase more steeply with date. Expected: newer-model slope greater than older-model slope. This is a conditional prediction, not a necessary consequence of leakage.

### Benchmark date versus next-token prediction loss



We froze a new random sample of **300 items, 30 per component**, before inference. Scoring used the same models, text preparation and primary 50% neutral setting as Experiment 1. There were 298 paired usable items at that boundary because two very short PIQA texts lacked a complete scorable suffix token.

**X-axis:** first arXiv submission date, spanning 2018-2020. This is a benchmark-age proxy, not a verified question creation or dataset release date. **Y-axis:** mean continuation loss in bits/byte. We averaged ARC Easy and Challenge into one family, giving nine equally weighted families for ordinary least-squares fits.

The expected-pattern panel is a schematic with arbitrary illustrative heights, not simulated or measured data. The observed panel uses the frozen results. MMLU is the only 2020 family, and every component predates the newer checkpoint.

## Experiment 2 / The observed loss slopes

Each point is a family mean. The two panels use the same y-axis range and include all nine families. The fitted lines summarize loss versus benchmark date; the question sample is not an official OLMES answer-accuracy evaluation.

**Actual family means: 30 items per OLMES component**



**Expected: a steeper increase for the newer model.**

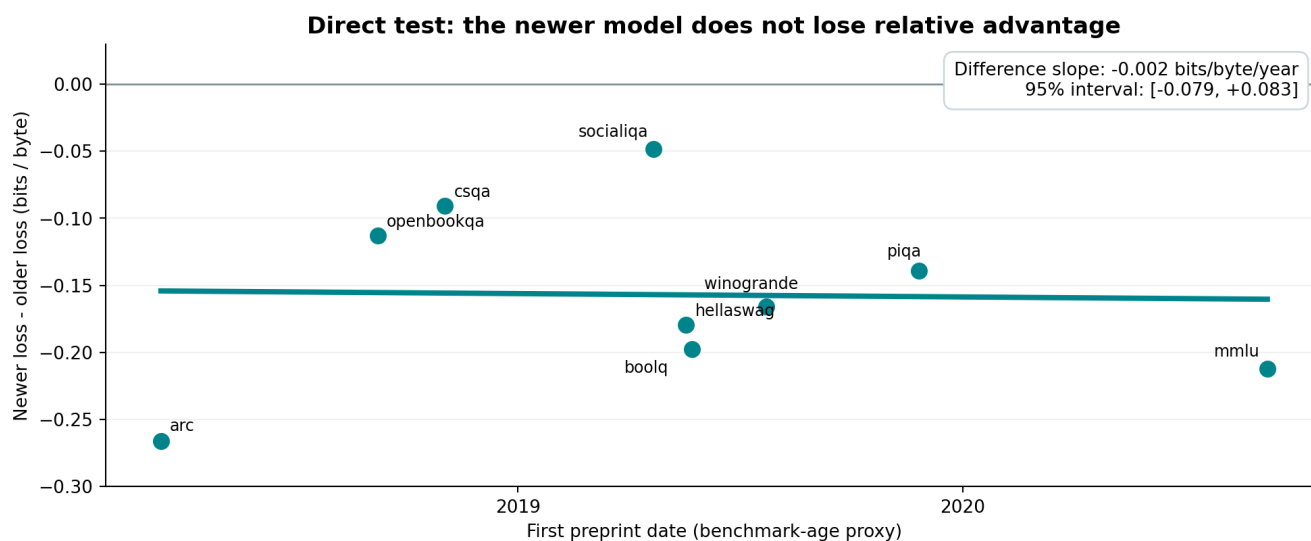
**Observed: +0.057 bits/byte/year for the older GPT-2 and +0.054 for the newer MiniCPM. The estimated slopes are nearly parallel.**

The newer model has lower mean loss on every family in this sample. Both fitted slopes are slightly positive, but the spread among tasks is large. PIQA and WinoGrande have high continuation loss for both models; the latest family, MMLU, has relatively low loss.

The positive fitted trends could reflect domain, wording or task composition. They do not establish a causal effect of benchmark age. The relevant test of our prediction is the difference between the slopes, shown next.

## Experiment 2 / Did the newer model degrade more?

For each family, we subtracted older-model loss from newer-model loss. A positive slope in this difference would support the predicted loss of relative advantage. Subtraction cancels only difficulty effects that are equal and additive for the two models; it does not remove every capability confound.



**Observed difference slope: -0.0025 bits/byte/year.**

**95% descriptive interval: [-0.079, +0.083].**

**We did not detect the predicted extra degradation in the newer model.**

The newer model retained about a 0.157 bits/byte advantage in both the 2018 and 2019-2020 groups. Both models worsened by about 0.079 between groups. The difference-in-differences was -0.0004 [-0.086, +0.105].

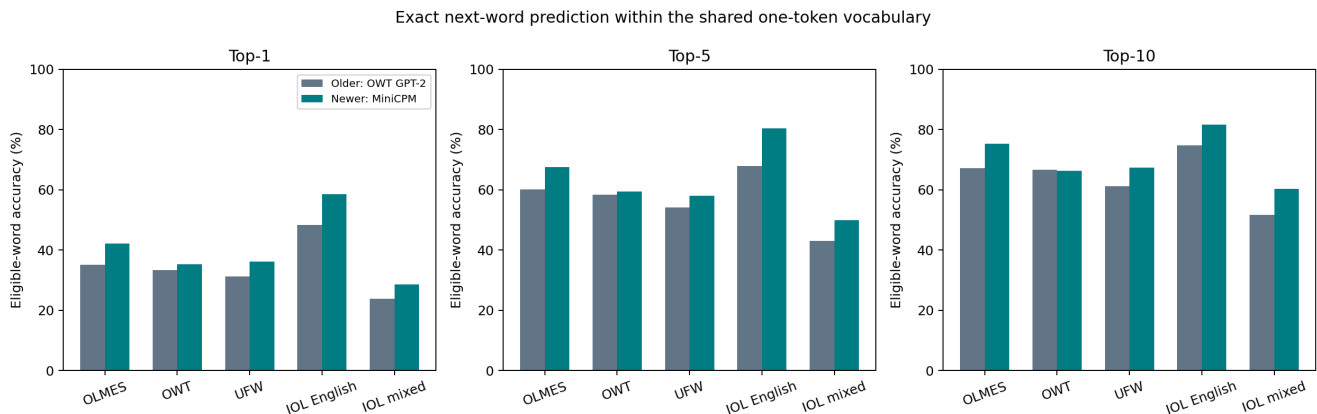
**Robustness:** removing MMLU gives a slope difference of +0.036 [-0.112, +0.107]. Leaving out each family in turn gives estimates from -0.055 to +0.036. All planned 25%/50%/75%, neutral/cued, raw/corpus-normalized and MMLU-exclusion intervals include zero.

Intervals use 10,000 family-bootstrap draws with fixed sampled-item means; they do not propagate all item-sampling or corpus-calibration uncertainty. Nine heterogeneous families provide limited precision. An interval spanning zero is not proof of equal slopes.

## Experiment 3 / Calibration-free word ranks

Raw negative log probability can differ across models because their probability scales differ. We therefore repeated the neutral 50% continuation comparison with exact next-word top-1, top-5 and top-10 accuracy. Words are divided only at literal ASCII spaces.

Both models still predict subword tokens. We restricted the candidate set and evaluated targets to **28,853 identical strings** that each tokenizer encodes as one canonical token: a leading space followed by one segment containing no whitespace. Multi-token words were excluded from both models rather than counted as misses. This gives identical eligible positions and a calibration-free ranking, with 80.2% coverage of OLMES continuation words.



**OLMES:** GPT-2 versus MiniCPM accuracy was 35.1% versus 42.2% at top-1, 60.1% versus 67.6% at top-5, and 67.2% versus 75.3% at top-10. The newer-model gains were therefore +7.1, +7.5 and +8.1 percentage points.

**Novel English control:** the newer-model gain was +10.3 points at top-1 and +12.6 at top-5, larger than its OLMES gain. General predictive ability remains a sufficient explanation. Relative to each model's own corpus control, the OLMES advantage rises by 4.3 top-1 points, 7.9 top-5 points and 7.4 top-10 points; the corpus text is not genre matched.

**Benchmark-date slopes of newer-minus-older accuracy: +0.017/year at top-1 [-0.029, +0.079], -0.010/year at top-5 [-0.036, +0.013], and -0.021/year at top-10 [-0.090, +0.029]. A negative slope matches the proposed loss of newer-model advantage on later benchmarks. Top-1 points the other way; every interval includes zero.**

# Combined interpretation and reproducibility

**Experiment 1:** OLMES was easier than mixed IOL text, but the mean ordering reversed for short English prose. This makes the apparent familiarity signal sensitive to the choice of control.

**Experiment 2:** the newer model did not show a steeper date-loss slope in this sample. **Experiment 3:** word ranks remove probability calibration, but top-1 and top-5/top-10 point in opposite temporal directions and all intervals include zero.

Together these results do not provide convincing evidence that leakage explains the observed differences. They also do not establish that either model is free from contamination. Loss can reflect language ability, repeated facts, templates, domain familiarity or direct exposure. No calibrated membership classifier or contamination-rate estimate was produced.

**A stronger follow-up: compare multiple years of the same exam series across a verified training cutoff, matching subject and format. Use confirmed consumed records where available and enough independent questions. Direct matches in the relevant training snapshot would strengthen exposure evidence. This follow-up has not been run.**

The original direct corpus-overlap proposal was superseded. Corpus samples supplied controls, but no full scan was completed. Hugging Face index-loading errors were not interpreted as evidence of absence.

## Execution and saved evidence

Inference used FP16 on a GTX 1650 (4 GB), batch size 1, evaluation mode, eager attention and no KV cache. No model or probe was trained. Final loss passes took 123.3 and 226.4 seconds. The final word-rank passes took 107.9 and 103.4 seconds. Cost: \$0 in paid services; timings exclude setup and earlier iterations.

Whitespace was normalized without changing case or punctuation. A common text budget allowed both tokenizers to stay below 500 tokens including cues. No pilot text was truncated; one long BoolQ text in Experiment 2 was shortened identically for both models. Both tokenizers preserved the inputs in round-trip checks. Five scoring tests passed. Temporal coverage and missingness were checked against token offsets and matched across models.

Frozen seeds: 20260911 for the pilot; 20260912 plus benchmark-name code-point sum for the temporal sample. Exact texts, hashes, row indices, revisions and scores are in `data/prediction/`, `data/temporal/`, `results/prediction/`, `results/temporal/` and `results/word_topk/`. Baseline corpus records and model-download manifests are retained. Bootstrap intervals remain exploratory.

Pinned GPT-2: 2f4e2079c9ac92c2b5c6fecc19fae645bcef56fa (2022-06-20). Pinned MiniCPM: f120ef901e3a11c2ed5fcfd242296cea3e80191b (2026-05-25). OLMES commit: 5a51f502d463b8cdc4a2dcad7d7096c41ff1197e.

Sources: [Stanford training/checkpoints](#); [MiniCPM checkpoint](#); [OLMES](#); [IOL 2026 problems](#). Each benchmark paper URL and first-submission date is saved in `data/temporal/manifest.json`.